

Dubrovin + Gamma

$$\frac{df(z)}{dz} = \frac{T}{z} f(z) \quad \text{sol. : } f(z) = e^{T \log z} := z^T$$

$$\frac{df}{dz} = \frac{T}{z^2} f \quad \text{sol. } f = e^{-T/z}$$

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

§1. Mirror symm. motiv:

§2. $\mathbb{P}^2$ (example for Dubrovin)

§3 Gamma conj.

References: Dubrovin ICM 1998

• Galkin - Golyshev - Iritani 2016

• Iritani ICM 2022

• Bayer - Manin 2001

• Cotti - Dubrovin - Guzzetti

① X Fano var. $(-K_X = c_1(X) > 0)$

two invariants:

• "A-model" $QH(X)$ as Frobenius manifold

• "B-model" $\mathcal{D}^b(Coh X) = \mathcal{D}^b(X)$

Dubrovin's conj. 1998:

(a) QH ^{sever.} semisimple $\Leftrightarrow$ $\mathcal{D}^b(X)$ has f.e.c. $\langle E_1, \dots, E_N \rangle$

(b) under cond (a), Stokes matrix $S = (s_{ij})$ equals (the inverse of) $\Leftrightarrow$ the Euler pairing matrix $\chi(E_i, E_j)$ for some exc. coll. $E_1, \dots, E_N$

(c) central connection C has the form $C = C' \cdot C''$ $C'' = (ch E_i)$
 $C' = \dots$

(later refined by Dubrovin to include Γ -class)

Mirror symmetry for X :

conjecturally $\exists W: X \rightarrow \mathbb{C}$
 regular function "LG model"

A-model of $X \approx$ B-model of W
 $h^{n,*} \left(\begin{array}{l} Fuk(X) \cong MF(W) \\ QH(X) \cong \text{"Saito"}(W) \end{array} \right) \rightarrow h^{n,*}$
 B-model of $X \approx$ A-model of W
 $D^b(X) \cong FS(W)$ (attribution)

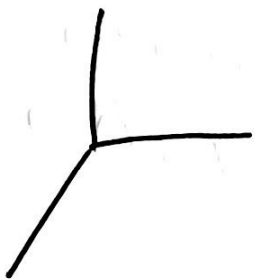
In this setting nondegenerate crit. pt of W plays special role and contribute as follows:

- A-model of W : Lefschetz thimble (vanishing cycles to a critical point)
(exc. object in $FS(W)$)
- Hence, $\mathcal{E} \in \mathcal{D}^b(X)$
- B-model of W : $MF(\bar{z}_1^2 + \dots + \bar{z}_n^2 = 0) = \mathbb{C}^*$
- hence: Lagr. cycle in X
and flat section of $QH(X)$ with good asymptotic properties

Particular case: "semisimple case",
i.e. W ~~semi~~ has all crit. pt nondegenerate:

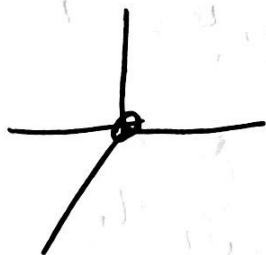
- A-model of W : FS has FEC of Lefschetz thimbles
- $\mathcal{D}^b(X)$ has FEC
- B-model of W : $MF(W) \simeq \prod \mathbb{C}^*$
- A-model of W : $Fuk(X)$ gener. by orth. Lagr.
- $QH(X)$ semisimple

Ex X toric Fano (e.g. $X = \mathbb{P}^n$)
 $W = \sum_{p \in \Sigma(1)} z^p$ (and appropriate
 q -def.) is LG



$$x + y + \frac{q}{xy}$$

$$\mathbb{P}^2$$



$$x + y + \frac{q_1}{xy} + q_2 xy$$

$$F_1 = \mathbb{P}^2$$

Note: X toric $\Rightarrow \mathcal{H}(X)$ f.e.c. $\checkmark$
 $\mathcal{Q}\mathcal{H}(X)$ semisimple $\checkmark$

Important remark / consequence of
 mirror symmetry: critical values
 of W = eigenvalues of the quant.
 mult. by $\mathbb{R} \cdot C_1(X)_q^*$.

Teveler, cartoon?

"semi simplicity"
in Mirror symm.

$\mathcal{L}(X)$ f.c.c. $\xleftrightarrow[\text{?}]{\text{Dolgovin}}$

$QH(X)$

semer. J.I.

take
 H^{odd}
(easy)

$$H^{odd}(X) = 0$$

$$H^{p,q}(X) = 0 \text{ for } p \neq q$$

all classes are H^{odd}

BM 2001
HMT 2008

Ex 2. Ex : $\mathbb{P}^2$ $\langle 0, 0(1), 0(2) \rangle$

$$\begin{pmatrix} 1 & 3 & 6 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 3 \\ & 1 & 3 \\ & & 1 \end{pmatrix}$$

$$\langle 0, \Omega(2), 0(1) \rangle$$

$$\Omega(2) = \text{Ker}(0^3(1) \rightarrow 0(2))$$

B-model

A-model

$$q = e^{t_2}$$

h^2 coord
small $\mathbb{Q}h$

$$c_1(X) \frac{*}{q} = \begin{pmatrix} 0 & 0 & 3q \\ 3 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$$

eigenvalues: $3\sqrt[3]{q}, 3\sqrt[3]{q}\omega, 3\sqrt[3]{q}\omega^2$

$\Rightarrow \mathbb{Q}h$ semisimple at $q \neq 0$

Stokes matrix?
 $\mathbb{Q}^n$

Delvarin $n=2$
Guzzetti any n

$$S_{ij} = \binom{n+1}{j-i} \quad i \leq j$$

$$S_{ij} = 0 \quad i > j$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

What is the Stokes matrix?

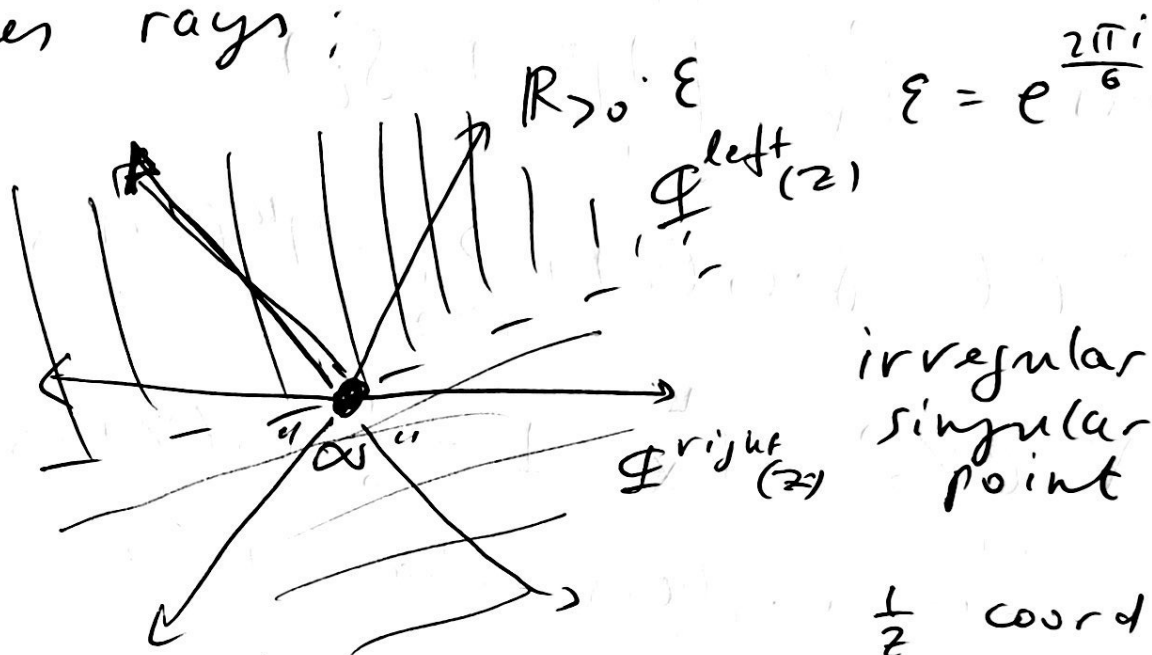
$$u_1 = 3\sqrt[3]{9}, \quad u_2 = 3\omega\sqrt[3]{9}, \quad u_3 = 3\omega^2\sqrt[3]{9}$$

eigenvalues

~ normalized idempotents

for quantum mult. π_1, π_2, π_3

Stokes rays;



$$\mathcal{I}_1 \sim \frac{1}{\sqrt{3}} \frac{e^{3z}}{z} (1 + O(\frac{1}{z})) \quad z \rightarrow \infty$$

$$\mathcal{I}_2 \sim \frac{\epsilon}{\sqrt{3}} \frac{e^{3\omega z}}{z} (1 + O(\frac{1}{z}))$$

$$\mathcal{I}_3 \sim \frac{\epsilon}{\sqrt{3}} \frac{e^{3\omega^2 z}}{z} (1 + O(\frac{1}{z}))$$

$z = c + i\omega$

$$g(z) = \frac{1}{(2\pi)^{1/2} i} \int_{c-i\infty}^{c+i\infty} \Gamma^3(-s) e^{\pi i s} z^{3s} ds$$

$$\Phi^{\text{right}}(z) = \begin{bmatrix} -g(\omega z) \\ g(z) \\ g(\bar{\omega} z) \end{bmatrix}$$

$S \Phi^{\text{right}} = \Phi^{\text{left}}$

$\Phi^{\text{left}}(z)$ similar

$$S = \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \sqrt{7}$$

5} Gamma-conjecture aim
to make the correspondence
more precise $\mathcal{J}^1(X) \hookrightarrow \mathcal{QH}(X)$

$$\Gamma(1-z)\Gamma(1+z) = \frac{\pi z}{\sin \pi z} \Rightarrow \hat{\Gamma}_X \cdot \hat{\Gamma}_X \sim t dx$$

$$\Gamma\text{-class: } \hat{\Gamma}_X = \prod_{i=1}^{\dim X} \Gamma(1+\delta_i) =$$

$$= \exp(-\gamma \cdot c_1(X) + \sum_{k \geq 2} (-1)^k (k-1)! \zeta(k) ch_k X)$$

e.g. $X \text{ (Y3) Libgober's}$ $\hat{\Gamma}_X = 1 - \frac{\pi^2}{6} c_2(X) - \zeta(3) (3D)$

(Kosono - Klemm - Theisen - Yau 1999)

$$K_0 X \xrightarrow{\hat{\Gamma}_X \cdot ch(-)} H^*(X, \mathbb{C})$$

$$Ch(E) = \sum_{k=0}^{2 \dim X} (2\pi i)^k ch_k(E)$$

$$\text{HRR } \chi(E_1, E_2) = \int ch E_1^\vee \cdot ch E_2 t dx$$

$$= [\hat{\Gamma}_X ch E_1, \hat{\Gamma}_X^\vee ch E_2]$$

$$[A, B] = \frac{1}{(2\pi)^{\dim X}} \int_X (e^{\pi i q(X)} e^{\pi i p} A) \cup B$$

Iritani

$KKP \Rightarrow \Gamma$ -dan used
to construct integral
structure on QH .

(matches mirror symm. pred.)

Gauss conjectures:

• Conjecture 0,

• Γ conj. I

• Γ conj. II

conj. 0 $C_1(X)_{\tau=0}^*$ has

the spectral radius $T = \max |\lambda|$
among eigenvalues, and it has
multipl. one.

known for many cases, incl. dp surf.
(slide)

Γ conj I X any Fano

$\hat{\Gamma}_X (= \hat{\Gamma}_X \cdot \text{Ch})$ transported
from the regular crit pt
to irregular one is the sol.
with the smallest asymptotic
 $\sim e^{-\text{Tr} z} \quad z \rightarrow +\infty$

Known for: dim 2, dim 3 $p=1$,
toric, hypers / c.i.

~~GGI~~ GGI: $\mathbb{P}^n, \text{Gr}(k, n)$

Semisimple case

Γ - conj II

A_i basis of

sols with prescribed asympt.
 $e^{-a_i/z}$ (in a sector, near irr. c.p.)

($a_i = e$ -values of ~~$c_1(X)_{\epsilon=0}^*$~~)

are given by $\hat{\Gamma}_X \cdot \text{Ch}(E_i)$ for

an exc. coll. $E_1 \dots E_m$ of $\mathcal{S}(X)$,

($\Rightarrow$ Dubrovin's conj. on Stokes
matrices)

Note Dubrovin reformulated (c)

C: columns $\frac{1}{(2\pi)^{d/2}} \hat{\Gamma}_X^- \cdot Ch E_i$
 $d = \dim X$

2027: Conj $\mathcal{O} + \Gamma$ conj I are
false for $\mathbb{P}_{\mathbb{P}^n}(\mathcal{O} \oplus \mathcal{O}(n))$ $n \geq 3$

~~possibly ray~~

e.g. $T \leftrightarrow \mathcal{O}$

largest
eigenvalue

look for other
 τ ?

[Galvin - Ku - Iritani - Ke - Li - Su]

Sanda - Shamoto: further
generalisation of Dubrovin + Γ
for Fano without exc. collection,
s.o.d. $\longleftrightarrow$, mutation systems for QK .
of solutions